

TEAM 02 - TIGERTRUCK DATABASE SCHEMA

Good Driver Incentive Program | 18 tables | 30 foreign-key relationships

Schema conventions

PK = Primary Key **FK** = Foreign Key **UQ** = Unique **NULL** = nullable. Composite PK/FK fields identify dependent child records. ABOUT_RELEASE is independent. Reports are generated from transactional data rather than stored as a separate table.

USER_ACCOUNT

Column	Type	Key / Constraint	Reference / Purpose
user_id	INT	PK	Unique user identifier
username	VARCHAR	UQ	Login username
password_hash	VARCHAR		Hashed password
first_name	VARCHAR		First name
last_name	VARCHAR		Last name
email	VARCHAR	UQ	Unique email address
phone	VARCHAR		Phone number
account_status	VARCHAR		Account status
created_at	DATETIME		Created timestamp
updated_at	DATETIME		Updated timestamp

ADMIN

Column	Type	Key / Constraint	Reference / Purpose
admin_id	INT	PK, FK	-> USER_ACCOUNT.user_id

SPONSOR_ORGANIZATION

Column	Type	Key / Constraint	Reference / Purpose
sponsor_id	INT	PK	Unique sponsor identifier
sponsor_name	VARCHAR		Organization name
status	VARCHAR		Sponsor status
point_dollar_rate	DECIMAL		Point-to-dollar conversion rate
contact_email	VARCHAR		Sponsor contact email
phone	VARCHAR		Sponsor phone

Column	Type	Key / Constraint	Reference / Purpose
created_at	DATETIME		Created timestamp

SPONSOR_USER

Column	Type	Key / Constraint	Reference / Purpose
sponsor_user_id	INT	PK, FK	-> USER_ACCOUNT.user_id
sponsor_id	INT	FK	-> SPONSOR_ORGANIZATION.sponsor_id
job_title	VARCHAR		Sponsor employee title

DRIVER

Column	Type	Key / Constraint	Reference / Purpose
driver_id	INT	PK, FK	-> USER_ACCOUNT.user_id
sponsor_id	INT	FK, NULL	-> SPONSOR_ORGANIZATION.sponsor_id
current_points	INT		Current point balance
participation_status	VARCHAR		Driver participation status
joined_at	DATETIME		Joined timestamp

DRIVER_APPLICATION

Column	Type	Key / Constraint	Reference / Purpose
application_id	INT	PK	Unique application identifier
driver_id	INT	FK	-> DRIVER.driver_id
sponsor_id	INT	FK	-> SPONSOR_ORGANIZATION.sponsor_id
decided_by_user_id	INT	FK	-> USER_ACCOUNT.user_id
submitted_at	DATETIME		Submission timestamp
status	VARCHAR		Application status
decision_at	DATETIME		Decision timestamp
reason	TEXT		Decision/rejection reason

POINT_TRANSACTION

Column	Type	Key / Constraint	Reference / Purpose
point_transaction_id	INT	PK	Unique point transaction
driver_id	INT	FK	-> DRIVER.driver_id
sponsor_id	INT	FK	-> SPONSOR_ORGANIZATION.sponsor_id
performed_by_user_id	INT	FK	-> USER_ACCOUNT.user_id
points_delta	INT		Points added/removed
balance_after	INT		Balance after transaction
reason	TEXT		Transaction reason
created_at	DATETIME		Created timestamp

RECURRING_POINT_SCHEDULE

Column	Type	Key / Constraint	Reference / Purpose
schedule_id	INT	PK	Unique schedule identifier
driver_id	INT	FK	-> DRIVER.driver_id
sponsor_id	INT	FK	-> SPONSOR_ORGANIZATION.sponsor_id
created_by_user_id	INT	FK	-> USER_ACCOUNT.user_id
points_amount	INT		Recurring point amount
reason	TEXT		Reason
frequency	VARCHAR		Schedule frequency
next_run_at	DATETIME		Next execution time
is_active	BOOLEAN		Schedule active flag

PRODUCT

Column	Type	Key / Constraint	Reference / Purpose
product_id	INT	PK	Unique product identifier
api_source	VARCHAR		External API/source
external_product_id	VARCHAR	UQ	External product identifier
product_name	VARCHAR		Product name
description	TEXT		Product description
dollar_price	DECIMAL		Dollar price
availability	VARCHAR		Availability/status
image_url	VARCHAR		Product image URL
category	VARCHAR		Product category
last_refreshed_at	DATETIME		Last API refresh

CATALOG_ITEM

Column	Type	Key / Constraint	Reference / Purpose
sponsor_id	INT	PK, FK	-> SPONSOR_ORGANIZATION.sponsor_id
product_id	INT	PK, FK	-> PRODUCT.product_id
point_price	INT		Sponsor point price
is_active	BOOLEAN		Catalog availability
added_at	DATETIME		Added timestamp

SHOPPING_CART

Column	Type	Key / Constraint	Reference / Purpose
cart_id	INT	PK	Unique cart identifier
driver_id	INT	FK	-> DRIVER.driver_id
sponsor_id	INT	FK	-> SPONSOR_ORGANIZATION.sponsor_id
status	VARCHAR		Cart status
created_at	DATETIME		Created timestamp
updated_at	DATETIME		Updated timestamp

CART_ITEM

Column	Type	Key / Constraint	Reference / Purpose
cart_id	INT	PK, FK	-> SHOPPING_CART.cart_id
sponsor_id	INT	PK, FK	part of -> CATALOG_ITEM
product_id	INT	PK, FK	part of -> CATALOG_ITEM
quantity	INT		Quantity
point_price_snapshot	INT		Point price at add time

CUSTOMER_ORDER

Column	Type	Key / Constraint	Reference / Purpose
order_id	INT	PK	Unique order identifier
driver_id	INT	FK	-> DRIVER.driver_id
sponsor_id	INT	FK	-> SPONSOR_ORGANIZATION.sponsor_id
placed_by_user_id	INT	FK	-> USER_ACCOUNT.user_id
status	VARCHAR		Order status
total_points	INT		Total points spent
total_dollar_amount	DECIMAL		Dollar-equivalent total
placed_at	DATETIME		Placed timestamp
cancelled_at	DATETIME		Cancellation timestamp

ORDER_ITEM

Column	Type	Key / Constraint	Reference / Purpose
order_id	INT	PK, FK	-> CUSTOMER_ORDER.order_id
line_number	INT	PK	Line number within order
sponsor_id	INT	FK	part of -> CATALOG_ITEM
product_id	INT	FK	part of -> CATALOG_ITEM
quantity	INT		Quantity
unit_point_price	INT		Point price snapshot
unit_dollar_price	DECIMAL		Dollar price snapshot

NOTIFICATION

Column	Type	Key / Constraint	Reference / Purpose
notification_id	INT	PK	Unique notification identifier
user_id	INT	FK	-> USER_ACCOUNT.user_id
notification_type	VARCHAR		Notification type
message	TEXT		Notification message
is_read	BOOLEAN		Read status
created_at	DATETIME		Created timestamp

ALERT_PREFERENCE

Column	Type	Key / Constraint	Reference / Purpose
user_id	INT	PK, FK	-> USER_ACCOUNT.user_id
point_change_enabled	BOOLEAN		Point-change alerts
order_summary_enabled	BOOLEAN		Order-summary alerts
drop_alert_enabled	BOOLEAN		Default TRUE

AUDIT_EVENT

Column	Type	Key / Constraint	Reference / Purpose
audit_event_id	INT	PK	Unique audit event
actor_user_id	INT	FK	-> USER_ACCOUNT.user_id
sponsor_id	INT	FK	-> SPONSOR_ORGANIZATION.sponsor_id
driver_id	INT	FK	-> DRIVER.driver_id
category	VARCHAR		Audit category
subject_username	VARCHAR		Affected username
entity_type	VARCHAR		Affected entity type
entity_id	VARCHAR		Affected entity identifier
success	BOOLEAN		Operation success flag
reason_or_details	TEXT		Reason/details
created_at	DATETIME		Created timestamp

ABOUT_RELEASE

Column	Type	Key / Constraint	Reference / Purpose
release_id	INT	PK	Unique release identifier
team_number	VARCHAR		Team number
version_number	VARCHAR		Release version
release_date	DATE		Release date
product_name	VARCHAR		Product name
product_description	TEXT		Product description

Foreign-Key Relationship Map

Child FK	Parent Key	Cardinality	Relationship
ADMIN.admin_id	USER_ACCOUNT.user_id	1:1	Identifying
SPONSOR_USER.sponsor_user_id	USER_ACCOUNT.user_id	1:1	Identifying
SPONSOR_USER.sponsor_id	SPONSOR_ORGANIZATION.sponsor_id	M:1	Non-identifying
DRIVER.driver_id	USER_ACCOUNT.user_id	1:1	Identifying
DRIVER.sponsor_id	SPONSOR_ORGANIZATION.sponsor_id	M:0..1	Non-identifying
DRIVER_APPLICATION.driver_id	DRIVER.driver_id	M:1	Non-identifying
DRIVER_APPLICATION.sponsor_id	SPONSOR_ORGANIZATION.sponsor_id	M:1	Non-identifying
DRIVER_APPLICATION.decided_by_user_id	USER_ACCOUNT.user_id	M:1	Non-identifying
POINT_TRANSACTION.driver_id	DRIVER.driver_id	M:1	Non-identifying
POINT_TRANSACTION.sponsor_id	SPONSOR_ORGANIZATION.sponsor_id	M:1	Non-identifying
POINT_TRANSACTION.performed_by_user_id	USER_ACCOUNT.user_id	M:1	Non-identifying
RECURRING_POINT_SCHEDULE.driver_id	DRIVER.driver_id	M:1	Non-identifying
RECURRING_POINT_SCHEDULE.sponsor_id	SPONSOR_ORGANIZATION.sponsor_id	M:1	Non-identifying
RECURRING_POINT_SCHEDULE.created_by_user_id	USER_ACCOUNT.user_id	M:1	Non-identifying
CATALOG_ITEM.sponsor_id	SPONSOR_ORGANIZATION.sponsor_id	M:1	Identifying
CATALOG_ITEM.product_id	PRODUCT.product_id	M:1	Identifying
SHOPPING_CART.driver_id	DRIVER.driver_id	M:1	Non-identifying
SHOPPING_CART.sponsor_id	SPONSOR_ORGANIZATION.sponsor_id	M:1	Non-identifying
CART_ITEM.cart_id	SHOPPING_CART.cart_id	M:1	Identifying
CART_ITEM.(sponsor_id, product_id)	CATALOG_ITEM.(sponsor_id, product_id)	M:1	Identifying
CUSTOMER_ORDER.driver_id	DRIVER.driver_id	M:1	Non-identifying
CUSTOMER_ORDER.sponsor_id	SPONSOR_ORGANIZATION.sponsor_id	M:1	Non-identifying
CUSTOMER_ORDER.placed_by_user_id	USER_ACCOUNT.user_id	M:1	Non-identifying
ORDER_ITEM.order_id	CUSTOMER_ORDER.order_id	M:1	Identifying
ORDER_ITEM.(sponsor_id, product_id)	CATALOG_ITEM.(sponsor_id, product_id)	M:1	Non-identifying
NOTIFICATION.user_id	USER_ACCOUNT.user_id	M:1	Non-identifying
ALERT_PREFERENCE.user_id	USER_ACCOUNT.user_id	1:1	Identifying
AUDIT_EVENT.actor_user_id	USER_ACCOUNT.user_id	M:1	Non-identifying
AUDIT_EVENT.sponsor_id	SPONSOR_ORGANIZATION.sponsor_id	M:1	Non-identifying
AUDIT_EVENT.driver_id	DRIVER.driver_id	M:1	Non-identifying

Relationship notation: 1:1 = one-to-one; M:1 = many child rows to one parent row; M:0..1 = many child rows may reference zero or one parent. Identifying relationships use the parent's key as part of the child's primary key; non-identifying relationships use a foreign key that is not part of the child's primary key.